

K-d Trees

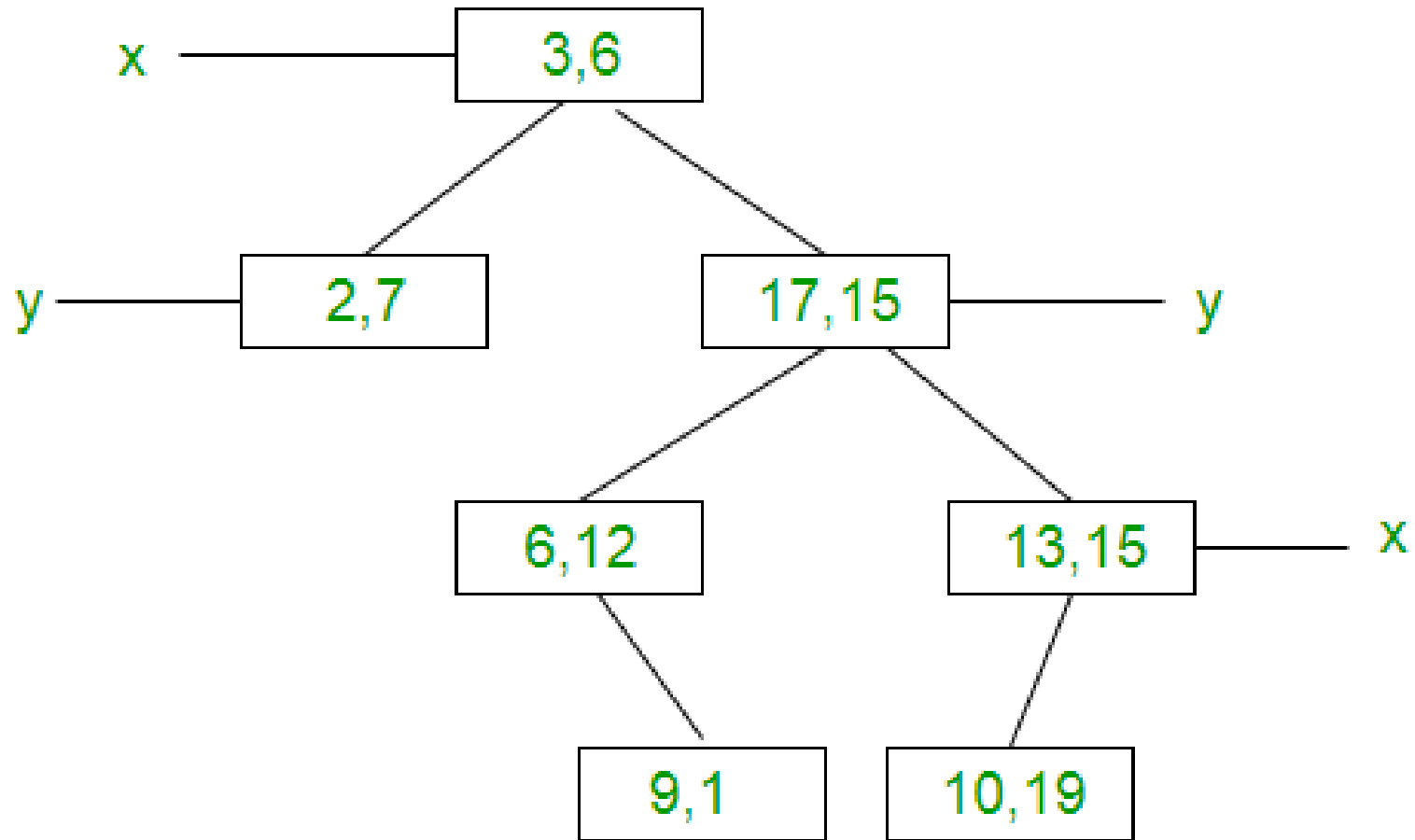
What is K dimension tree?

- A K-D Tree(also called as K-Dimensional Tree) is a binary search tree where data in each node is a K-Dimensional point in space.
- In short, it is a space partitioning(details below) data structure for organizing points in a K-Dimensional space.
- A non-leaf node in K-D tree divides the space into two parts, called as half-spaces.
- Points to the left of this space are represented by the left subtree of that node and points to the right of the space are represented by the right subtree.

What is K dimension tree?

- **Creation of a 2-D Tree:** Consider following points in a 2-D plane: (3, 6), (17, 15), (13, 15), (6, 12), (9, 1), (2, 7), (10, 19)
 1. Insert (3, 6): Since tree is empty, make it the root node.
 2. Insert (17, 15): Compare it with root node point. Since root node is X-aligned, the X-coordinate value will be compared to determine if it lies in the right subtree or in the left subtree. This point will be Y-aligned.
 3. Insert (13, 15): X-value of this point is greater than X-value of point in root node. So, this will lie in the right subtree of (3, 6). Again Compare Y-value of this point with the Y-value of point (17, 15) (Why?). Since, they are equal, this point will lie in the right subtree of (17, 15). This point will be X-aligned.
 4. Insert (6, 12): X-value of this point is greater than X-value of point in root node. So, this will lie in the right subtree of (3, 6). Again Compare Y-value of this point with the Y-value of point (17, 15) (Why?). Since, $12 < 15$, this point will lie in the left subtree of (17, 15). This point will be X-aligned.
 5. Insert (9, 1): Similarly, this point will lie in the right of (6, 12).
 6. Insert (2, 7): Similarly, this point will lie in the left of (3, 6).
 7. Insert (10, 19): Similarly, this point will lie in the left of (13, 15).

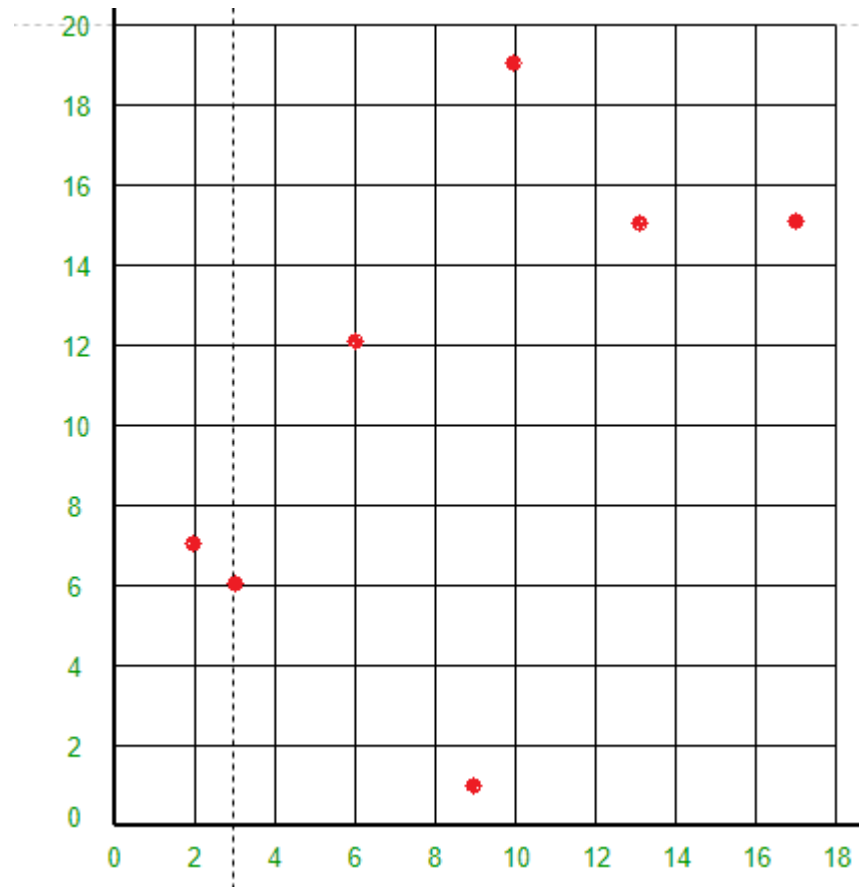
What is K dimension tree?



How is space partitioned?

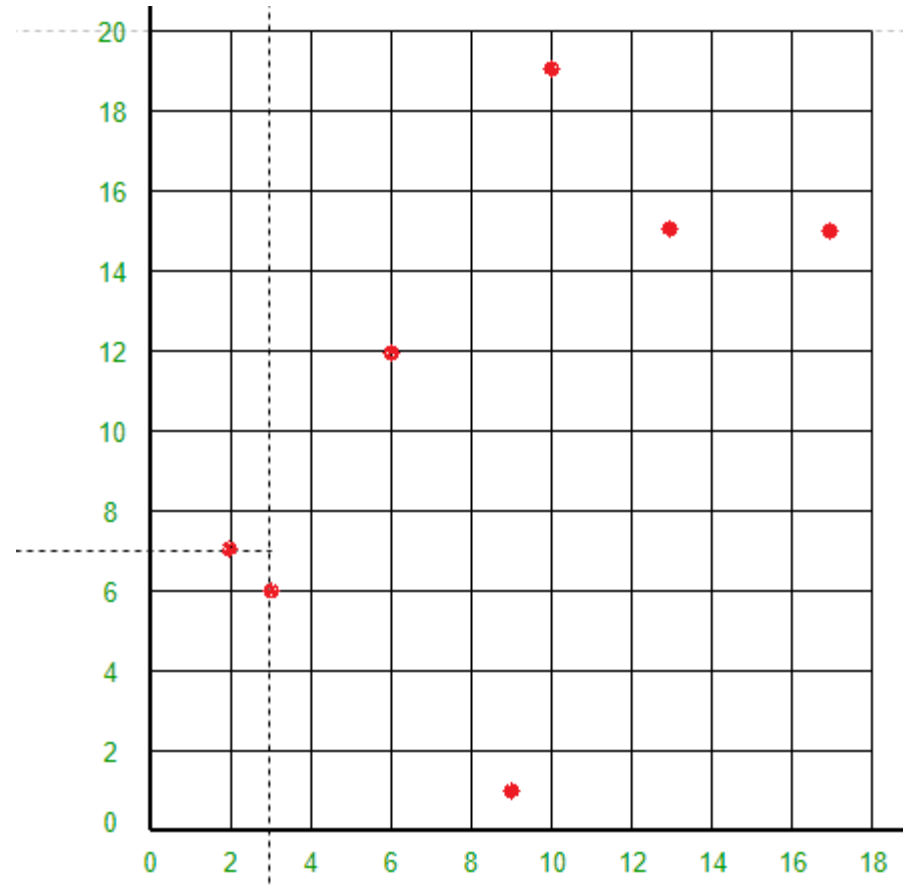
All 7 points will be plotted in the X-Y plane as follows:

Point (3, 6) will divide the space into two parts: Draw line $X = 3$.



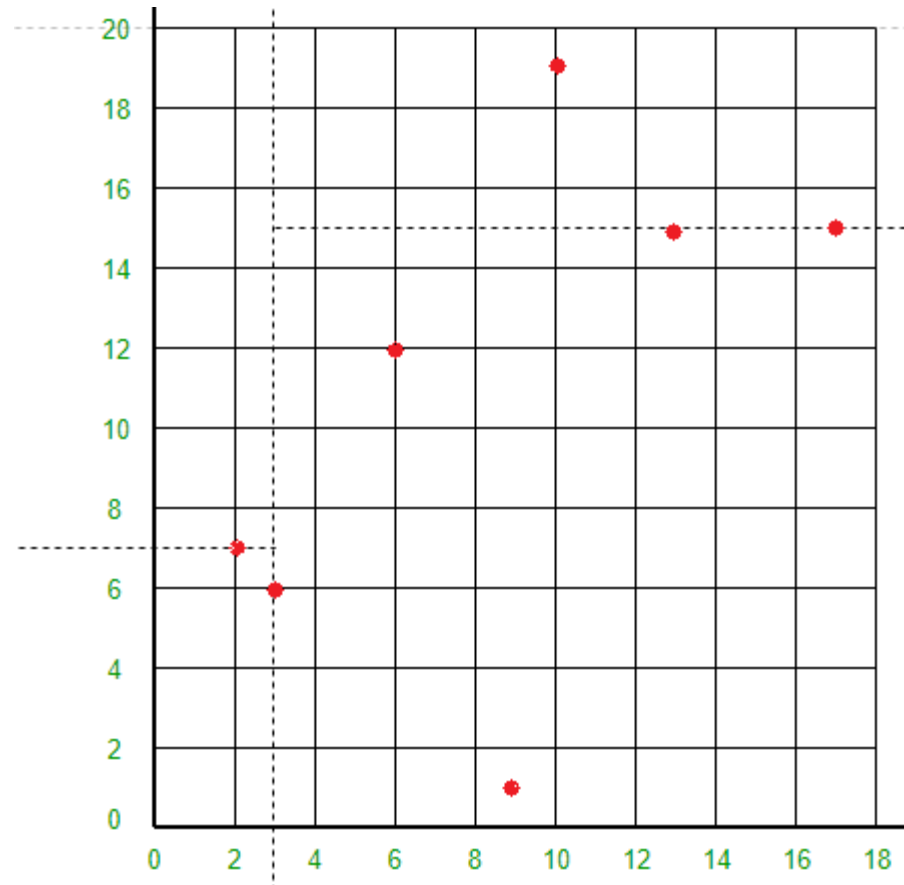
How is space partitioned?

- Point (2, 7) will divide the space to the left of line $X = 3$ into two parts horizontally. Draw line $Y = 7$ to the left of line $X = 3$.



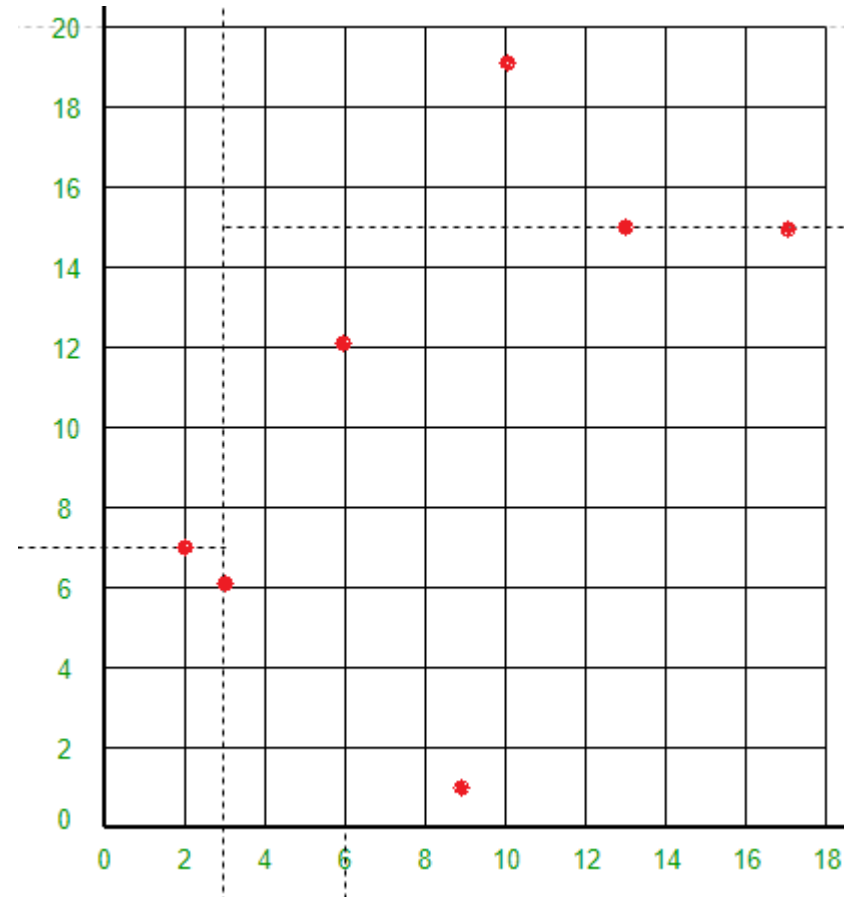
How is space partitioned?

- Point (17, 15) will divide the space to the right of line $X = 3$ into two parts horizontally. Draw line $Y = 15$ to the right of line $X = 3$.



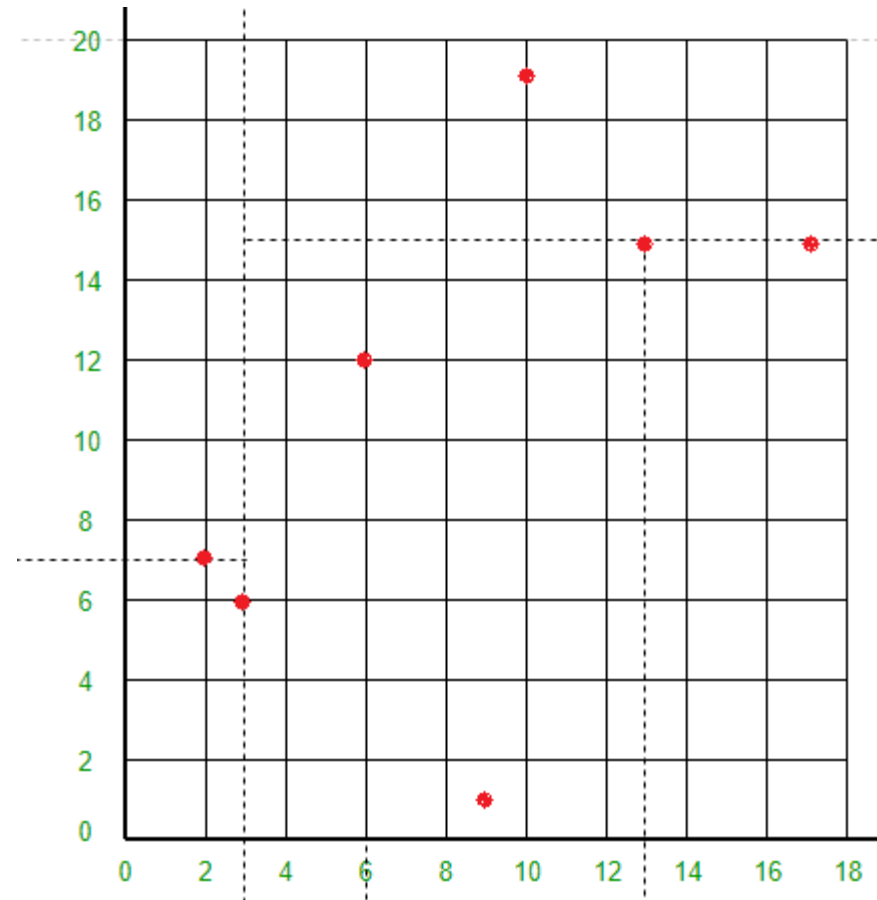
How is space partitioned?

- Point (6, 12) will divide the space below line $Y = 15$ and to the right of line $X = 3$ into two parts. Draw line $X = 6$ to the right of line $X = 3$ and below line $Y = 15$.



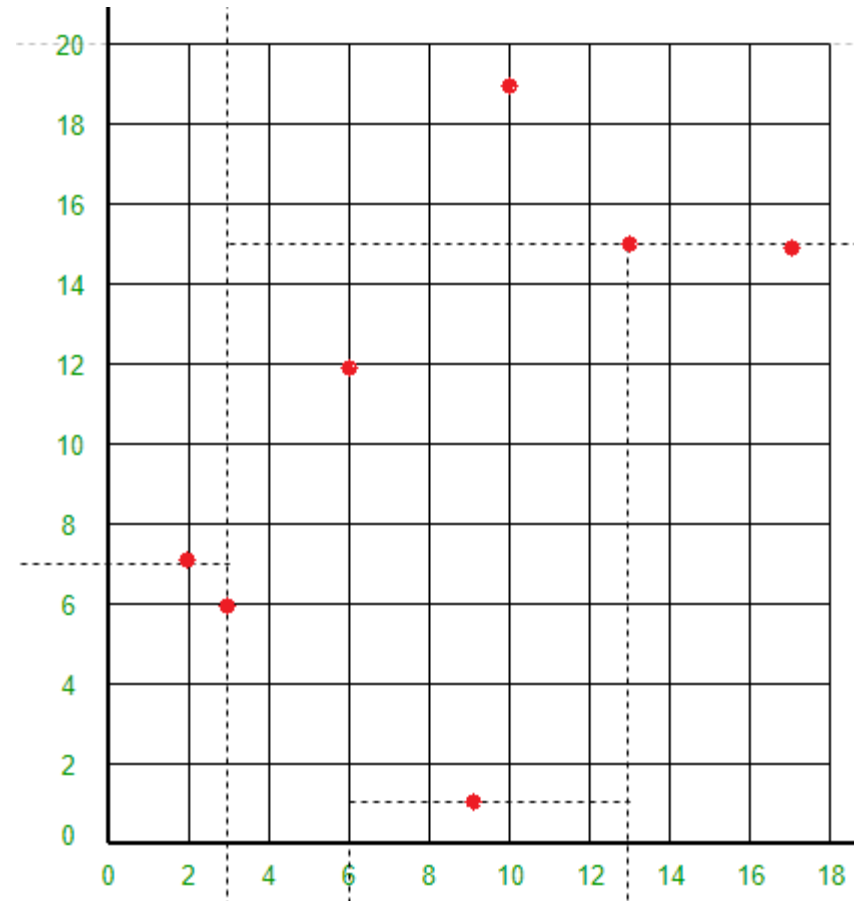
How is space partitioned?

- Point (13, 15) will divide the space below line $Y = 15$ and to the right of line $X = 6$ into two parts. Draw line $X = 13$ to the right of line $X = 6$ and below line $Y = 15$.



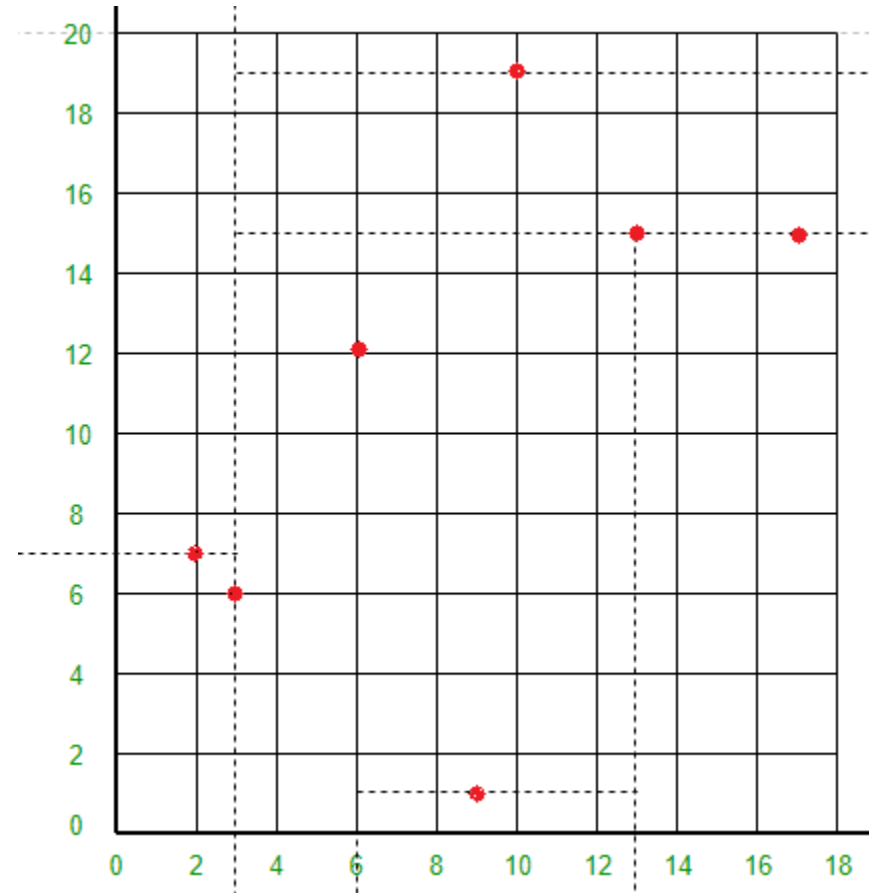
How is space partitioned?

- Point (9, 1) will divide the space between lines $X = 3$, $X = 6$ and $Y = 15$ into two parts. Draw line $Y = 1$ between lines $X = 3$ and $X = 13$.



How is space partitioned?

- Point (10, 19) will divide the space to the right of line $X = 3$ and above line $Y = 15$ into two parts. Draw line $Y = 19$ to the right of line $X = 3$ and above line $Y = 15$.



Reference

- <https://www.geeksforgeeks.org/search-and-insertion-in-k-dimensional-tree/>